User involvement in healthcare technology development and assessment

Structured literature review

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Abstract

Purpose – Medical device users are one of the principal medical device technology stakeholders. The involvement of users in medical device technology development and assessment is central to meet their needs. This study aims to examine this issue.

Design/methodology/approach – A structured review of the literature published from 1980 to 2005 in peer-reviewed journals was carried out from a social science perspective to investigate user involvement practice in the development and assessment of medical device technologies. This was followed by a qualitative thematic analysis.

Findings – Medical device users include clinicians, patients, carers and others. Different kinds of medical devices are developed and assessed by user involvement. The user involvement occurs at different stages of the medical device technology lifecycle and the degree of user involvement is in the order of: design > testing and trials > deployment > concept stages. The methods most commonly used for capturing users' perspectives are usability tests, interviews and questionnaire surveys.

Research limitations/implications – The relevant engineering, medical and nursing literature, which might have been useful, was not reviewed. However, useful findings emerge that apply to health care generally.

Originality/value – This study shows that medical device users are not homogeneous but heterogeneous in several aspects, such as needs, skills and working environments. This is an important consideration for incorporating users' perspectives in medical device technologies.

Keywords User studies, Medical equipment, Design and development

Paper type Literature review

Introduction

Medical device users are one of medical device technology's primary stakeholders. Therefore, knowledge of their needs and their involvement in medical device development and assessment (MDD&A) is important. Devices that meet the needs of users enhance safety (Kaye, 2000; Chiu *et al.*, 2004), while non-consideration of user needs has serious consequences (Stone and McCloy, 2004), for example the occurrence

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of medical device errors (Amoore and Ingram, 2002; Food and Drug Administration, 2003; Baker *et al.*, 2004; Bennett *et al.*, 2005), which are important from the users' view point (Samore *et al.*, 2004). Additionally, understanding users' needs is important as it determines the success or failure of technology development (Cahill *et al.*, 1994; Shaw, 1998) and the quality of the product (Keiser and Smith, 1994). Developing better products requires in-depth consideration of all users and their activities, their actual daily working environment, and their functional limitations, innumeracy and skills (Ostrander, 1984; Wilkins and Holley, 1998; Rockwell, 1999; Green *et al.*, 2000; Staccini *et al.*, 2001; Kittel *et al.*, 2002; Kaufman *et al.*, 2003). If user requirements affect all aspects of device development, then acquiring them has to be done properly (Tsai *et al.*, 1997) and involving users in technology development and assessment results in the production of more successful medical devices (Brockhoff, 2003, Biemans, 1991; Shaw, 1998; Lin *et al.*, 2001).

The aim of this literature review is to investigate the practice of user involvement in MDD&A as reported within the social science literature. In particular, we wanted to answer the following questions:

- What kinds of medical devices are developed and assessed by user involvement?
- What types of medical device users are involved in MDD&A?
- What is the extent of user involvement at different stages of the medical device technology lifecycle?
- What methods are used for capturing user perspectives in MDD&A?

Method

The Multidisciplinary Assessment of Technology Centre for Healthcare (MATCH) is a collaboration between five leading UK universities in healthcare technology assessment along with a cohort of industrial partners. It is sponsored by the UK Engineering and Physical Sciences Research Council (EPSRC) to support the healthcare technology sector and its user communities by creating methods to assess value from concept through to mature product and by engaging with regulatory bodies at home and abroad (Multidisciplinary Assessment of Technology Centre for Healthcare, 2005). One programme project is targeted at engaging users. It aims to develop formal methods for evaluating users' perspectives and to engage the community (Multidisciplinary Assessment of Technology Centre for Healthcare, 2003). The project comprises three teams, i.e.:

(1) healthcare;

(2) engineering and ergonomics; and

(3) social science.

The authors, part of the social science team, conducted an extensive structured review of literature published mainly in social sciences journals and conference proceedings from January 2004 to April 2005. Literature in healthcare and engineering and ergonomics disciplines was not surveyed by the authors since it was reviewed by the healthcare team (Bridgelal Ram *et al.*, 2005) and the engineering and ergonomics team (Martin *et al.*, 2005), respectively. However, a few studies from the other disciplines might have been included in this study owing to their availability on bibliographic databases searched by the authors. Consequently, Mulholland *et al.* (2000), Dutt *et al.*

(2002), Garmer *et al.* (2002a, 2004) and Liljegren and Osvalder (2004) have been omitted from our analysis since they have been reported by Bridgelal Ram *et al.* (2005) and Martin *et al.* (2005). The literature review process was adapted from Beverley *et al.* (2004) and Bruce and Mollison (2004). Key words included:

- device users;
- end-users;
- medical device;
- medical device users;
- needs assessment;
- new medical technology;
- user centred product;
- user criteria;
- user input;
- user interests;
- user involvement;
- user needs;
- user needs assessment;
- user needs research;
- user participation;
- user perceptions;
- user perspective;
- user requirements;
- user requirements elicitation;
- user studies; and
- user survey.

Searches were conducted across the following online bibliographic databases:

- Blackwell Synergy;
- Ebscohost;
- Emerald Insight;
- International Bibliography of the Social Sciences (IBBS);
- IEEE/IEE Electronic Library;
- Ingenta;
- JSTOR;
- Kluweronline;
- Medical Device Link;
- ProQuest;
- Sage Publications;

- ScienceDirect;
- Social Science Information Gateway (Sosig); and
- SpringerLink.

Criteria for article selection included studies that reported user involvement in MDD&A published from 1980 to 2005 in English. Studies where there was no user involvement during any stage of the medical device lifecycle were excluded from the review. Three reviewers – a full time research fellow and two social science PhD students (interns for a three-month period) – reviewed the selected articles. A data extraction template (Figure 1) was developed in-house in spreadsheet format. For data extraction purpose, the medical device lifecycle was divided into five stages (Table I):

(1) concept;
(2) design;
(3) testing and trials;
(4) production; and
(5) deployment.

These stages were based on the product lifecycle reported in the literature (Cooper and Kleinschmidt, 1986; Rochford and Rudelius, 1997; World Health Organization, 2003).

						Product development stages at which users were involved							
ID	Reference	Year & Country of Study	Objective(s)/Question(s)	Device/Product	Method/Tool/Approach	Users/Participants	Concept	Design	Testing & Trials	Production	Deployment	Findings &/or Conclusion	Researcher's Comments
1	2	3	4	5	6	7	8a	8b	8c	8d	8e	9	10

Figure 1.
Data extraction template

Table I.
Stages of the product lifecycle

Stage	Details
Concept	Starts with idea generation and includes technical, financial and commercial assessment
Design	Involves product development process from (re)design to prototype development
Testing and trials	Starts with prototype testing in house and includes trails in the real field
Production	Includes production on large scale supported by business and commercial rationale
Deployment – marketing, launch and use	Includes product marketing, launch and use in the real field

Article identification, short-listing, reviewing and data abstraction comprised three phases:

(1) searching, using terms and quick reading of titles and abstracts that identifyied relevant articles;
(2) careful abstract reading that identified and short listed the most promising articles; and
(3) thoroughly reading short-listed articles followed by extracting to the template.

After the abstraction process was completed, data were cleaned and cross-checked; this process showed that 28 articles were reviewed more than once. The data abstracted from these articles by different reviewers were compared and no significant difference was found when reviewers had processed the same article. This confirmed the reliability of the abstracted data and helped to validate the process.

Descriptive statistics and qualitative thematic analysis (Ritchie and Spencer, 1994) were used for analysing the data, which were divided into different themes, i.e.:

- types of medical devices developed and assessed;
- types of medical device users involved;
- extent of user involvement by different stages of the medical device development cycle; and
- methods used for capturing users' perspectives.

The articles reviewed were mainly Bandolier evidence taxonomy type V (Bandolier, 1994; Hurst, 2005). A few studies also used type IV evidence. The total number of participants (ranging from two to 370) and breakdown of their characteristics were clearly mentioned in 54 per cent ($n = 13$) of the studies reviewed.

Results

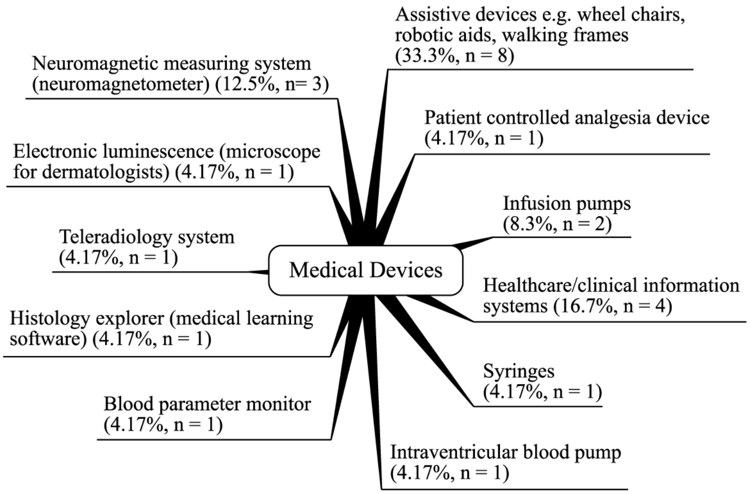
We reviewed the social sciences literature and elicited user perspectives on medical device development and assessment. Other products were also reviewed that have us a broader picture. Here we present data from 24 studies (see the Appendix) that reported user involvement in the development and assessment of medical device technologies. The findings from our qualitative thematic analysis along with the studies' descriptive statistics follows.

Types of medical devices developed and assessed by user invovlement

The review revealed that a variety of medical devices (Figure 2) was developed and assessed by involving medical device technology users. Devices ranged from very simple medical equipment such as syringes to complex ones such as a neuromagnetometer, which is used to analyse and diagnose human brain cortex activity and problems (Hasu, 2000; Hasu and Engestrom, 2000; Miettinen and Hasu, 2002), and included the devices that are used by various types of users.

Types of medical device users involved in medical device development and assessment

A wide range of users was involved in the MDD&A process, including clinicians, patients, carers, family members and persons with different disabilities and impairments (Figure 3).



Note: % = frequency, n = number, total studies = 24

Figure 2.
Types of medical devices developed and assessed by user involvement

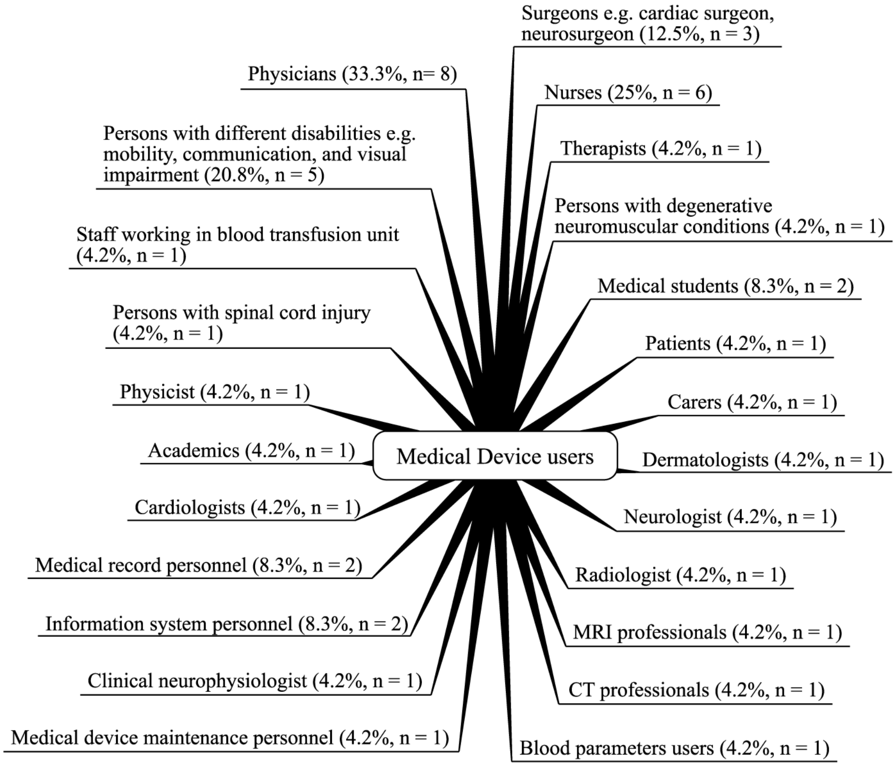
Extent of user involvement by stage of the medical device lifecycle

In 50 per cent ($n = 12$) of the studies, users were involved in one stage of the medical device lifecycle and in the remaining studies (50 per cent, $n = 12$) users were involved in more than one stage of the lifecycle. Single-stage involvement was highest in the deployment phase in 21 per cent ($n = 5$) of the studies, followed by design stages in 17 per cent ($n = 4$). Testing and trials stage in 8 per cent ($n = 2$) and concept stage in 4 per cent ($n = 1$) of total studies. Two-stage combinations were between design and testing and trials; concept and design stages; and testing and trails and deployment stages. The user involvement in each of the two stage combinations was in 8 per cent ($n = 2$) of studies. Three-stage combinations for user involvement were between concept, design, and testing and trials stages in 13 per cent ($n = 3$) followed by design, testing and trials, and deployment stages in 4 per cent ($n = 1$) of studies. Four-stage user involvement was between concept, design, testing and trials, and deployment stages in 8 per cent ($n = 2$) of studies. The extent of overall user involvement in each stage of medical device technology lifecycle was highest in the design stage (58 per cent, $n = 14$) and lowest in the concept stage (33 per cent, $n = 8$), which is shown in Figure 4.

Methods used for capturing users' perspectives

Involving users and capturing their perspectives in the medical device technology lifecycle concerned mostly usability tests in 33 per cent ($n = 8$), interviews in 25 per cent ($n = 6$), questionnaire surveys in 21 per cent ($n = 5$), discussions in 8 per cent ($n = 2$), simulations in 8 per cent ($n = 2$), and design sessions, focus groups, Delphi method, human factors approach, observation, task analysis, use experiment, user and producer seminars, user feedback, and video recording each in 4 per cent ($n = 1$) of the studies. These methods are mapped against the medical device lifecycle stages where they were used (Table II). Some of the studies used more than one method of inquiry.

Usability tests, interviews, questionnaire surveys and user-producer seminars were used in all four medical device lifecycle stages where the users were involved.



Notes: % = frequency, n = number, total studies = 24. Some studies involved more than one type of the users

Figure 3.
Types of medical device users involved in medical device development and assessment

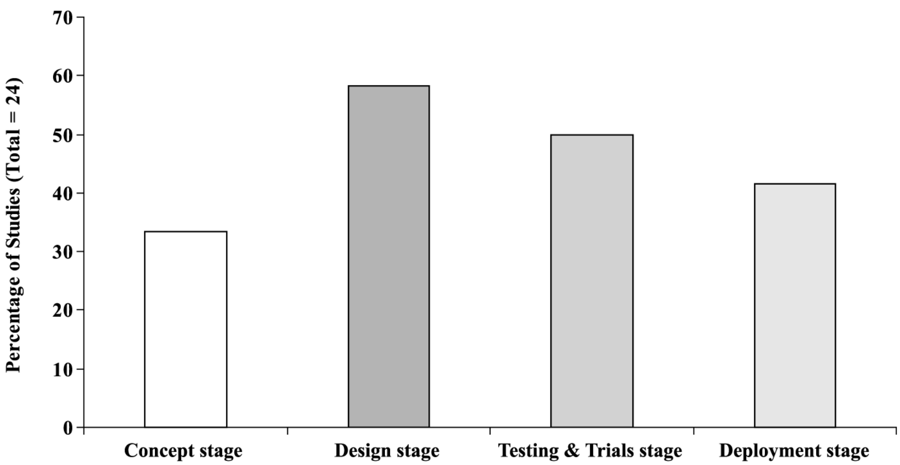


Figure 4.
Extent of user involvement by stage of the medical device lifecycle

Concept stage	Design stage	Tests and trials stages	Deployment stage
Interviews	Interviews	Interviews	Interviews
Usability tests	Usability tests	Usability tests	Usability tests
Questionnaire surveys	Questionnaire surveys	Questionnaire surveys	Questionnaire surveys
User and producer seminars	User and producer seminars	User and producer seminars	User and producer seminars
Task analysis	Task analysis	Task analysis	Use experiment
Discussion	Discussion	Video recording	Video recording
Observations	Observations	Observations	Focus groups (Delphi method)
Simulations	Simulations	Simulations	
Users' feedback	Human factors approach	Human factors approach	
	Design sessions		

Table II. Methods used for capturing user perspectives by stages of the medical device lifecycle

However, usability tests, interviews and questionnaire surveys were most commonly used for involving users and capturing their perspectives at various stages of the medical device lifecycle. Interviews were mainly semi-structured and face-to-face. Other methods used at one or two stages of the lifecycle included:

- design sessions during the design stage;
- focus groups during the deployment stage; and
- "discussions" during the concept and design stages.

Discussion

Our review reveals that various types of medical devices are developed and assessed by users. These devices, based on their use, mainly include assistive, surgical and administration equipment. These, however, are not the only medical devices that can be developed and assessed by involving users. There might be other types of medical devices that are developed and assessed by user involvement but were not identified in this review owing to the nature of databases selected.

Medical devices are divided into different classes according to the European Commission (1993, 2001) and the USA classifications (Food and Drug Administration, 1997). Information regarding the class of medical apparatus identified in this review could not be ascertained from the studies reviewed. It is therefore not possible to say that certain classes of medical devices can be developed and assessed by involving users. The review also shows that there are different groups of healthcare technology users. Users' characteristics, skills and working environment may therefore be different, which is worth considering when developing health care technologies from the users' perspective. The review shows that different types of medical device users can be involved in device development and assessment. However, we note that some users are less available, including the elderly and the disabled (Marshall *et al.*, 2002) and clinical consultants, particularly in highly sought-after clinical specialities such as cardiac surgeons or consultants working in accident and emergency departments. In this case, substitutes called "user surrogates" may be involved. For instance, physician surrogates working in emergency departments in joint application development sessions seemed acceptable since the surrogates had worked with physicians and were aware of physicians' needs (De and Ferratt, 1998a, b). Decisions regarding user surrogates must be made after advantages and disadvantages are considered (Bradley *et al.*, 1983; Herbert and Salmon, 1994; Cook and Woods, 1996; Gotzsche *et al.*, 1996; Tsevat *et al.*, 1998; Moinpour *et al.*, 2000). Also, MDD&A user involvement is dependent on regulatory approval since there are ethical and legal issues involved (McGregor and Brophy, 2005).

None of the studies reviewed mentioned the financial implications of involving users. Neither was there mention of financial remuneration given to users for participating in the research. However, this does not imply that there are no costs associated with users, who might participate on a volunteer basis. There might be other costs involved in the MDD&A user involvement process which need to be studied. We noted that users are involved in all stages of the medical device technology lifecycle except the development stage. This may be because this stage relates to full-scale manufacturing supported by a business and commercial rationale. Kaulio (1998) identified three interfaces of user involvement:

(1) specification;

(2) concept development; and

(3) prototyping – new product development where actual user involvement takes place during the design stage.

Gyula (2001) found that user involvement was higher in the initial and marketing stages. However, we found that user involvement occurs during four of the five stages of medical device technology lifecycle – concept, design, testing and trials, and deployment stages. The extent of user involvement by stage shows that the highest level was during the design followed by testing and trials, deployment and concept stages. What is common between our review and studies by Kaulio (1998) and Gyula (2001) is higher user involvement during the early than in the latter medical device technology lifecycle stages, which may be because user involvement in the early stages saves time and costs including those associated with later modifications (Tsai *et al.*, 1997; Giuntini, 2000; McDonagh *et al.*, 2002). User involvement during the early stages of development lifecycle is regarded as important (Sato and Salvador, 1999; Truffer, 2003) as it helps to adopt user needs (Saiedian and Dale, 2000). Therefore, user involvement at early stages has been suggested (Kyng, 1991; Ornetzeder, 2001) because it benefits both users and producers (Sanford *et al.*, 1998). User involvement at the design stage was highest because their contribution is considered vital (Sanford *et al.*, 1998; Dorup *et al.*, 2001). Highest user involvement at the design stage helps develop user-centred designs – essential to product success (Wai and Siu, 2003). The resultant designs are used to develop medical devices that have higher market usability (Gould and Lewis, 1985), improved equipment safety and efficiency (Lin, 1998) and those that may be successfully used and maintained (Fouladinejad and Roberts, 1996). Another reason for higher user involvement at the design stage may be the regulatory requirement of formal design processes starting and ending with customer needs (Powers and Greenberg, 1999).

Our review shows that numerous methods, both direct (e.g. interviews, usability tests and questionnaire surveys) and indirect methods (e.g. observation and simulation) (Noyes and Starr, 1995) are designed to involve users in the development and assessment of healthcare technologies. All enquiry methods have advantages and disadvantages: however, some methods are more appropriate and their selection and application depends on the purpose and context of the inquiry. For example, focus groups are useful for product concept evaluation (McQuarrie and McIntyre, 1986) and involving users in the interface design process (Nielsen, 1997). Similarly, the user feedback method is regarded important for user involvement throughout medical device development process such as blood monitoring (Bray, 2000). The development of better products requires in-depth understanding of all types of users, their activities and needs (Staccini *et al.*, 2001). However, involvement of different types of users, understanding their needs and eliciting their perspectives for developing medical devices require a combination of both qualitative and quantitative methods (Biemans, 1991; Edwards and Staniszewska, 2000; Tenopir, 2003). Combining usability tests and contextual inquiry in early stages of the product lifecycle, according to Rockwell (1999), leads to better targeted products, higher customer satisfaction and reduced development time. Combining methods also helps us to understand the usability aspects of medical devices (Garmer *et al.*, 2004), for example:

- using the human factor approach and usability tests for redesigning a volumetric infusion device (Garmer *et al.*, 2002b);
- usability tests, structured interviews and discussions for capturing user requirements in assistive technology (Buhler, 1996);
- questionnaire survey and usability tests (Lacey and Slevin, 2001) and task analysis, observation, simulation and questionnaire survey (Green *et al.*, 2000) for developing user-centred designs;
- theoretical activity perspective and usability tests for new medical technology (Hasu, 2000);
- focus groups and usability tests for ventilator development (Garmer *et al.*, 2004); and
- interviews and usability tests for developing and assessing infusion devices (Obradovich and Woods, 1996).

The selection of methods for capturing users' perspectives, however, depends on the type of medical device. Hyysalo (2003) believes that eliciting user needs for radically innovative technologies is yet to be proven by traditional methods such as market surveys and expert interviews. Emerging healthcare technologies require user involvement in trials, while established healthcare technologies require discussions with experienced and long-term users (Buhler, 1996). Since capturing user perspectives at various stages of medical device lifecycle depends on the method applied, selection of an appropriate method is important. Additionally, selecting appropriate method(s), selecting users and the mode and timing of their involvement are critical in medical device technology development and assessment.

Conclusion

Involving medical device users at different stages of the medical device lifecycle is important for developing and assessing healthcare technologies from users' perspectives. Users are mainly involved during the medical device lifecycle design and testing and trial stages. Involving users and eliciting their perspectives may require careful selection of appropriate method(s). Future research therefore needs to investigate the benefits and barriers associated with user involvement in the development and assessment of medical device technologies.

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Appendix: Studies that reported user involvement in medical device development and assessment

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